

AI-Powered Badminton Coach: Automated Match Analysis and Key Event Annotation

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Abstract—This research presents an AI-powered automated badminton match analysis and annotation system. Our system processes match videos to identify key actions, determine stroke types, track shuttlecock trajectories, and recognize point winners. To achieve this, we employ computer vision and deep learning techniques [1] to detect player joint positions and track the shuttlecock's trajectory. Given the inherent challenges in tracking shuttlecock movements, such as motion blur, occlusion, and small object size, we also use temporal data analysis techniques to enhance detection accuracy. Based on the preprocessed data, we systematically determine the hitting player and classify shot trajectories, ultimately generating a structured timeline of match events. The primary contributions of this work include a fully automated shot and rally annotation framework, reducing reliance on manual labeling, and providing structured performance insights for players and coaches. Additionally, it enhances spectator engagement by offering intuitive match visualizations. Our research provides a way to automatically extract and label shot types and rally sequences from raw match videos, bridging the gap between AI-driven sports analysis and practical applications.

Index Terms—Badminton analysis, Event detection, Shot classification, Computer vision, Sports analytics

I. INTRODUCTION

Sports analytics has experienced a significant transformation with the integration of computer vision methods and deep learning models, enabling automatic extraction of information from video footage. These technologies allow for real-time tracking, event detection, and tactical analysis, offering valuable feedback for players, coaches, and analysts. Among these sports, analytics of badminton competitions present unique challenges due to rapid shuttlecock movement, frequent occlusions or visual confusion with background, and complex player interactions. All of the above challenges make automated analysis a demanding task.

Previous research has explored various aspects of badminton match analysis, including stroke forecasting, player performance assessment, and object tracking [11]. While many existing approaches rely heavily on manually labeled datasets or focus primarily on predictive modeling, automated annotation remains underexplored. In this work, we introduce an AI-powered badminton match analysis system that aims to enhance rally segmentation, shot classification, and match event annotation. By integrating computer vision and temporal smoothing techniques, our approach extracts

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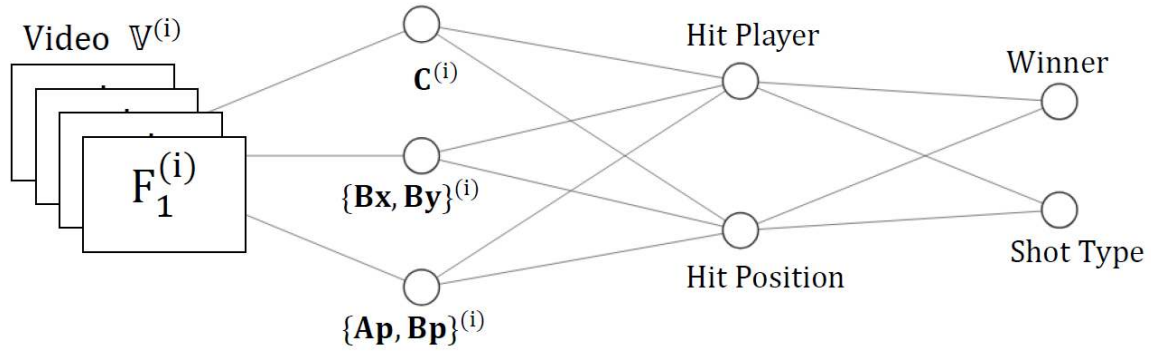


Fig. 1: Overall badminton competition analysis flow chart

structured insights from match videos, improving the accuracy and efficiency of badminton analytics while reducing reliance on extensive manual labeling. Our validating and testing badminton competition videos are mainly from ShuttleSet22 [3], [4]. By applying our method, we can capture the video's shuttlecock trajectories, hit positions, and shot styles. Based on these detections, the system provides real-time analysis of strategies, shot accuracy, and scoring, offering a robust framework for evaluating player performance and competitive dynamics (Figure 1). This not only assists players and coaches in monitoring gameplay and refining strategies but also improves spectator engagement by offering intuitive match visualizations, making the game more accessible to a broader audience.¹ Additionally, our framework can be reformulated into a neural network-like structure, allowing for flexible integration with existing neural network architectures and methods, thereby expanding its applicability in AI-driven sports analytics.

II. RELATED WORKS

Computer vision, supported by machine learning and deep learning models, is transforming sports analytics by enabling the automatic extraction of insights from video footage. This technology can detect objects [18], track movements [17], and analyze player performance, providing valuable data for strategy development and performance improvement. Systems such as those in [14] have shown success in ball trajectory tracking, while models like [15] [17] [16] focus on applying multiple targets tracking to the sport video for tracking multiple players' movement. In addition, with the use of pre-trained networks like YOLO [18] [19], models for motion tracking can be further fine-tuned and offer a high accuracy for analytics. Similarly, machine learning models have been employed for predictive analytics, such as match outcome prediction [20]. These advancements provide the foundation for modern AI-driven sports analytics frameworks, though challenges remain in automating complex event detection in fast-paced games like badminton.

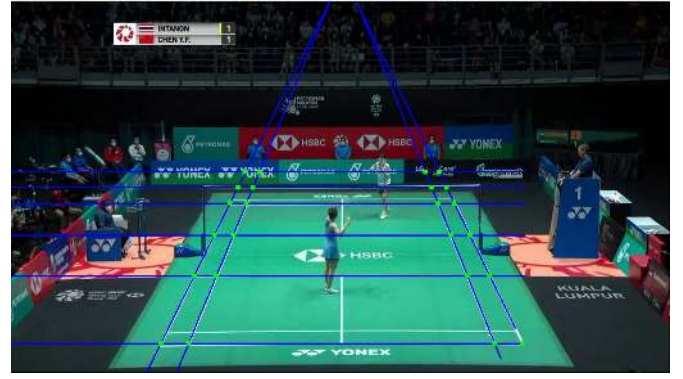
Considering fast-paced games such as badminton, tennis, and table tennis, tracking high-speed objects is an important and significant technical challenge. Due to the shuttlecock's rapid acceleration and unpredictable motion, traditional tracking methods often fail under occlusion and motion blur. Recent studies have employed deep learning-based tracking models to improve accuracy [14] [23]. Specifically, Huang et al. introduced TrackNet [22], a deep neural network designed for tracking fast-moving small objects. In addition, TrackNetV2 [1] was introduced in the next year. TrackNetV2 employs a Fully Convolutional Network (FCN) to transform input frames into heatmaps, indicating the desired object's probable locations. By incorporating temporal information across consecutive frames, TrackNetV2 predicts object trajectories, even when the object is temporarily occluded. Compared to the previous version, TrackNetV2 rapidly improves processing speed and performance, providing a baseline for the ball-tracking modeling development [24].

Other than detecting players and the ball in the competition, separating shots in a rally is critical in sports analytics. Here, a rally refers to the exchange of shots in a badminton competition while the shuttlecock is in play. In fast-paced games like badminton and tennis, techniques that involve identifying individual shots within a continuous sequence of play are essential for analyzing player performance, strategy development, and game outcome prediction. As these techniques become increasingly mature, integrated systems combining multiple methods to provide comprehensive information have been developed. Wang et al. introduced Coach AI [21], a deep learning-based system designed for badminton match analysis. This system effectively extracts key information, including player movements, positioning, and strategic patterns, while also tracking shuttlecock trajectories to identify swing moments, rally segmentation, shot categorization, and point outcomes. By providing structured match insights, Coach AI is a valuable tool for coaches and athletes, assisting in strategic planning and performance optimization. In addition, Hsu et al. [9] developed a system that integrates shuttlecock tracking and hit detection using a single camera to improve the analysis of badminton games. In addition, they employed a hit detection

¹Demo Website: <https://tu211.github.io/badminton/demo.html>



(a) Initial Hough line detection (horizontal)



(b) Updated court line after automated pruning

Fig. 2: **Court Line Detection** Applying Canny edge detection [6] and Hough Line Transform [7] provides us with an initial set of detected lines. (a) gives an example of the horizontal group of lines for processing. After our pruning process, the final court line result is shown in (b), with green dots denoted as the final court boundary points.

algorithm that identifies the moments when players strike the shuttlecock, allowing the system to segment the game into individual rallies based on these detected hits.

Our work further enhances automated badminton analytics by integrating computer vision with temporal smoothing techniques to improve rally segmentation and shot classification accuracy. Our framework builds upon prior research with Openpose [2], TrackNetV2 [1], and temporal data analysis methods such as Autoregressive Integrated Moving Average model (ARIMA) [5] to provide all the information from the temporary information of ball, players, and the rally segmentation, shot classification, and further the rally winner. Unlike existing systems that primarily focus on feature extraction and manual interpretation, our approach aims to develop a fully automated framework, minimizing reliance on manual annotations while delivering structured and scalable match analysis and clear information for players and viewers.

III. METHODS

A. Dataset Description

The video data utilized in this work for analysis is sourced from the **ShuttleSet22** dataset. ShuttleSet22 is an extension dataset from ShuttleSet [3] with videos from the badminton competitions in 2022. It consists of video recordings and corresponding match records from Badminton World Federation (BWF) singles matches. In addition, the video is segmented to capture a rally from the serve to the shuttle drop, and the detailed event annotation of the recorded data is provided. Detailed dataset attributes can be found in CoachAI's GitHub repository.

In our analysis, we exclusively use BWF singles match videos from the dataset. These videos are captured from a single-camera perspective, using an overhead view similar to standard badminton broadcasts. While this top-down perspective offers a comprehensive view of the court, it introduces specific challenges, such as reduced visibility of players farther from the camera and perspective distortion affecting court

geometry. These videos serve as the primary input for our AI-based badminton analysis system, providing valuable visual data for shot classification, rally segmentation, and player movement tracking. The event annotations serve only as a reference during training and refinement. By relying solely on video data, our method enhances automation in rally segmentation and shot classification without requiring extensive manual annotations. Furthermore, our framework can be reformulated into a neural network-like structure, enabling seamless integration with existing deep learning models for more advanced badminton analytics.

B. Data Preprocessing and Feature Extraction

In order to generate the features used for the following automated detection, we first split the feature extraction task into three subtasks: detecting the court, the shuttlecock trajectories, and the players' body keypoints. To describe the process with a theoretical way (Fig. 1), we consider a badminton video $\mathbb{V}^{(i)}$ as our input data, with each frame denoted as

$$\{F_j^{(i)}, j = 1, \dots, m^{(i)}\}$$

To accurately detect the court lines and corner coordinates, we first apply Canny edge detection [6] and Hough Line Transform [7] to generate an initial set of detected lines. Based on their coordinates and orientation, we refine this set by filtering out irrelevant lines and retaining only the most probable candidates. Since the initial detection often includes numerous redundant or inaccurate lines, we further classify them into horizontal and vertical groups and process them separately (Fig. 2(a)). Extraneous lines outside the court boundaries are removed, and the remaining lines are clustered based on their distances. This clustering process is performed twice to ensure a stable and accurate set of court lines, denoted as $\mathbf{C}^{(i)}$. Although this filtering process helps prune most court lines, some may still appear misaligned or incomplete due to camera distortions. To handle this, we apply perspective transformation, which corrects the detected court structure by compensating for the camera angle and converting the distorted

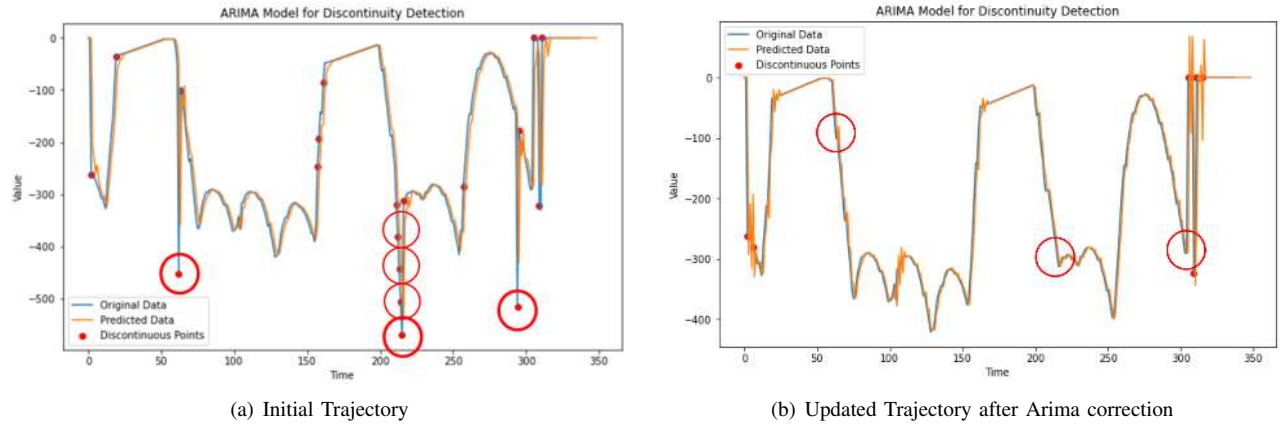


Fig. 3: **Anomaly Trajectory Correction based on Arima** The red points inside the red circle indicate the anomaly points detected by Arima. In (a), the value even went out of the frame dimension. From (b), we can observed that the values become more reasonable, and the red circles indicate all the calibrating results based on the interpolation.

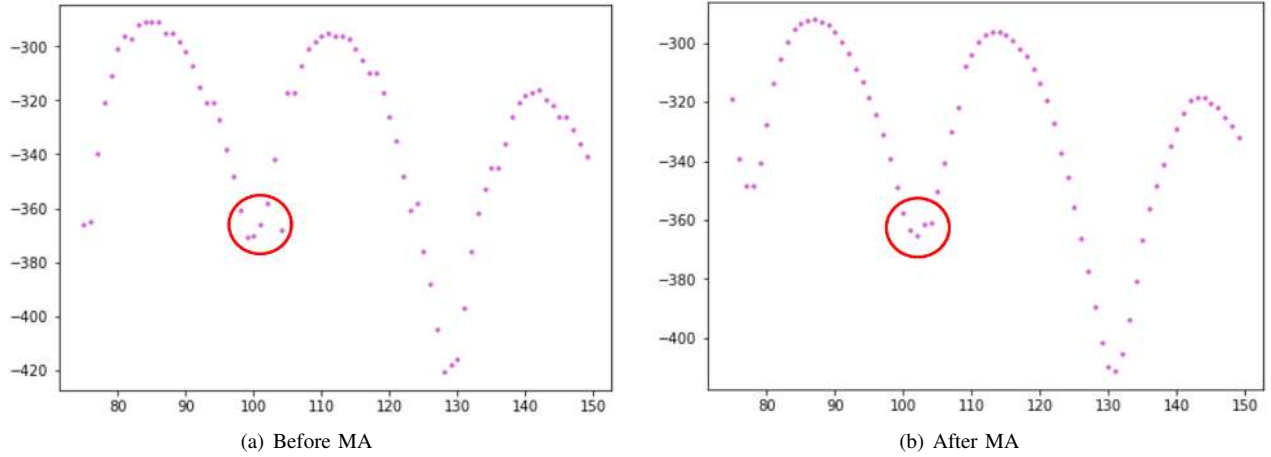


Fig. 4: **Shuttlecock trajectory smoothing by MA**

court shape into a properly aligned rectangular representation. The final court coordinates are obtained by comparing results from both the filtering and transformation steps. Points with minimal coordinate discrepancies are marked as valid, and the set with the highest number of valid points is selected as the final court boundary representation. (see Fig. 2 as an example of court line detection and transformed result).

In addition to the court detection, our framework also detects the shuttlecock trajectories and the players' body keypoints from the time series data $\{F_j^{(i)}\}$. We first apply TrackNetV2 developed by Sun et al. [1] to each frame $F_j^{(i)}$ and generate the initial position $(x_j, y_j)^{(i)}$ of the shuttlecock. However, the trajectory can be bumpy or irregular due to the shuttlecock being blocked by players or the net in the video, or players wearing white-color shoes or shirts. To solve the false detection, we first apply the autoregressive integrated moving average (ARIMA) [5] to find the anomaly points and recalculate the coordinates based on interpolation from the previous and the next coordinates (Fig. 3.). Moreover,

to eliminate the bias from the data itself, we also apply another moving average (MA) additionally to the data in order to smooth the whole trajectory (Fig. 4.). This way, we generate the final shuttlecock coordinate $\{\mathbf{Bx}, \mathbf{By}\}^{(i)}$ based on the input data $\{F_j^{(i)}\}$. We also apply a similar smoothing process to the players' initial 25 body keypoints generated by OpenPose [2], which is developed by CMU - Panoptic Studio. This way, we could generate the final keypoints of players $\{\mathbf{Ap}, \mathbf{Bp}\}^{(i)}$ (here we denoted the player at the upper court as A , the one at the lower court as B .)

With the above preprocessing and feature extraction process, we create a set of 170 features, including the shuttlecock trajectory, players' body keypoints, and the keypoints coordinates of the court, along with the distances between the shuttlecock and players' hands (considering as the distance of shuttlecock and each player.)

XGBoost [8]	SVM	Perceptron	Decision Tree	Logistic Regression	Random Forests	Bagging
88.58%	95.31%	89.09%	88.58%	95.41%	87.36%	94.9%

TABLE I: Accuracy of detecting which player hits the shuttlecock with different machine learning methods and the final resulting accuracy by using bagging.

C. Advanced Feature Extraction

In addition to the 170 basic features generated in the previous section, we apply machine learning methods to detect who hit the shuttlecock and where it was shot. To achieve this, we employ an ensemble learning approach, integrating multiple machine learning models through bootstrap aggregation. To determine the hitting player, we adopt a majority voting scheme among six machine learning models, where Player A is assigned as 0 and Player B as 1. If more than three models (out of six) predict the same player, that prediction is directly accepted. However, a weighted decision is applied when the models reach a 3:3 tie. Specifically, the individual model accuracies are summed, and their average serves as a threshold. The accuracy sum of the three models predicting Player B is compared against this threshold—if it exceeds the threshold, the hitter is assigned as Player B (1); otherwise, it is Player A (0). This weighted decision ensures a higher confidence level when standard majority voting is inconclusive. Our method achieves a 94.9% accuracy in identifying the hitting player (Table I), surpassing the performance of individual models such as SVM and logistic regression regarding overall confidence and robustness.

Beyond player detection, our method also identifies the exact hitting frame using player prediction results and shuttlecock trajectory $\{\mathbf{Bx}, \mathbf{By}\}^{(i)}$ analysis. The hitting frames are first estimated by detecting boundaries between consecutive shots, using the predicted hitter results as segmentation markers. Observing that each hitting segment contains at least one hitting event, we use these boundaries as reference intervals to refine the detection process. Within each segment, we analyze the shuttlecock angle variations to further pinpoint the precise hitting moment. The shuttlecock angle is measured relative to a vertical baseline, ranging from 0° (bottom) to 180° (peak). A significant increase in angle serves as a strong indicator of a hitting event. Initial predictions may contain false positives due to model misclassification, particularly in the first segment, where the shuttle may not be immediately detected due to the player gripping the shuttlecock before serving. This situation can lead to false large-angle variations. To solve this issue, false angle detections and false removal must be performed. Similarly, shuttle bounces after landing in the final segment may introduce misclassified hits. Thus, any detections within this interval are removed. Lastly, for segments where no hitting frame is initially found, the angle threshold is adaptively lowered until at least one hitting event is detected. After refinement, the detected hitting frames maintain an average deviation of less than 5 frames (0.2 seconds at 30 fps), ensuring precise event segmentation and improving shot classification accuracy.

D. Winner Detection and Shot Type Detection

Based on the shuttlecock trajectory $\mathbf{Bx}, \mathbf{By}^{(i)}$, court information $\mathbf{C}^{(i)}$, and hitting details, we determine the winner of each rally by analyzing the landing position and trajectory characteristics of the shuttlecock. Observing the final stroke in each video, we categorize shuttle trajectories into two primary types and apply a case-based winner prediction method based on the trajectory's slope.

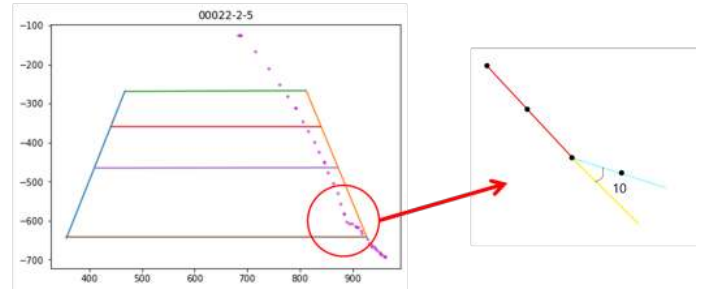


Fig. 5: The first type of trajectory (flat motion)

The first type of trajectory occurs when a player hits the shuttlecock forward in a relatively flat motion. This is typically seen in drives or smashes, where the angle of change remains small. If the shuttle deviates more than 10 degrees from its original path and does not continue along its expected route, it is classified as having landed, after which we verify whether it was in or out of bounds (Fig. 6.)

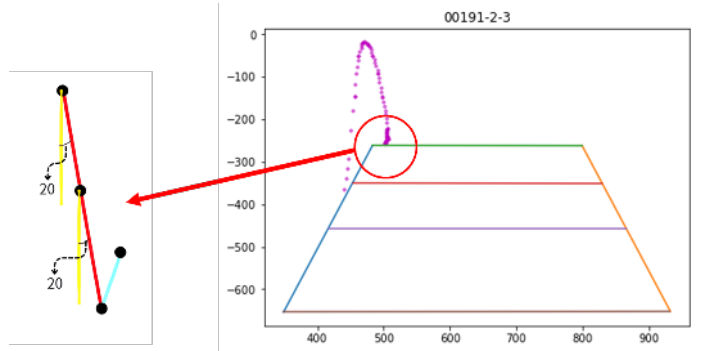


Fig. 6: The Second type of trajectory (bouncing motion)

In contrast, the second type follows a high-arching motion, where the shuttle is lifted upwards until reaching its peak, at which point its velocity and kinetic energy approach zero before descending in a parabolic trajectory. This type is common in clears or lifts and exhibits a more significant variation in trajectory angle. If the angle remains consistently below 20 degrees, followed by a sharp change, the shuttle is

TABLE II: Features Used for Shot Type Classification

Feature Description	Number of Variables
Player joint coordinates (x, y, z) for both players	75
Court boundary coordinates (x, y)	16
Shuttlecock movement distance	1
Shuttlecock speed	1
Shuttlecock angle variation (x, y)	2
Shuttlecock trajectory segmented into 3 parts (distance and angle change per segment)	6
Foot y-axis movement relative to service line	4
Foot x-axis movement relative to center line	4
Joint movement variations over previous 5 frames	18
Elbow angle variations of the hitting player (x, y)	2
Hitting player identification	1
Cross-court shot indicator (yes/no)	1
Total	131

also considered landed, and its final position is examined for boundary violations.

Several factors, such as trajectory anomalies or occlusions caused by players, may interfere with the winner detection process in both cases. After filtering out these cases, we evaluated 203 rallies, successfully determining the winner in 187 cases, with 16 incorrect classifications. The primary source of errors was the shuttle passing beneath the net, leading to misclassification. The method achieved an overall accuracy of 92.12%, with trajectory type one (flat trajectory) achieving 88.73% accuracy across 71 rallies, and trajectory type two (parabolic trajectory) reaching 93.94% accuracy across 132 rallies.

Our framework also provides shot type detection. The original ShuttleSet dataset [3] provides 18 different shot types. Here we consider the most common 8 type: short service (0), long service (1), clear (2), drive (3), smash (4), net shot (5), drop (6), lob (7). After testing multiple machine learning methods, here we use logistic regression to detect each shot type, and the overall accuracy falls around 73% (Table IV.). From the confusion matrix in Figure 7, we can see that drive and drop perform worse than other types. This may due to the less amount of data and the similar action between two types.

Our framework also provides shot type detection based on a combination of shuttlecock trajectory, player joint positions, and court information. The original ShuttleSet dataset [3] defines 18 different shot types, but we focus on the 8 most common types, as listed in Table III. To enhance classification accuracy, we extract a set of comprehensive features derived from shuttlecock movement, court boundaries, and player joint positions. The detailed of all the features we extracted are list in Table II.

TABLE III: Shot Types Considered in Our Model

Shot Type Index	Shot Type Description
0	Short Serve
1	Long Serve
2	Clear
3	Drive
4	Smash
5	Net Shot
6	Drop
7	Lob

We trained four different machine learning models for classification: Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost [8]. The training and testing accuracy of each model is summarized in Table IV.

TABLE IV: Model Performance for Shot Type Classification

Classifier	Training Accuracy	Testing Accuracy
Logistic Regression	0.82	0.73
SVM	0.78	0.70
Random Forest	0.80	0.70
XGBoost	0.77	0.71

Based on these results, we selected Logistic Regression as our final classifier due to its relatively higher test accuracy and stability. However, analyzing individual shot types reveals that drive (3) and drop (6) exhibit lower accuracy, as seen in Table V.

TABLE V: Per-Class Accuracy of Shot Type Classification

Shot Type	Accuracy
Short Serve (0)	0.82
Long Serve (1)	0.69
Clear (2)	0.61
Drive (3)	0.40
Smash (4)	0.80
Net Shot (5)	0.85
Drop (6)	0.51
Lob (7)	0.70

From the confusion matrix (Figure 7), we observe that drive and drop shots have lower classification accuracy, which may be due to the limited number of training samples and the similar motion patterns between these shot types. This suggests that additional feature engineering or larger datasets may further improve classification performance. Overall, our approach integrates various detection methods, extracts a comprehensive set of features, and employs multiple machine learning models to enhance accuracy. While most aspects achieve high precision, shot classification remains a challenging task, and hitting frame prediction still requires refinement. Nevertheless, our system successfully integrates the predicted data with match videos, allowing viewers to analyze badminton matches from an alternative, data-driven perspective (Fig. 8.) This aligns with our goal of leveraging AI to provide a new way to interpret and understand badminton gameplay, paving the way for further advancements in automated sports analytics.

0	192	5	0	2	5	23	3	2
1	3	38	1	0	0	0	1	12
2	0	2	131	3	4	31	13	30
3	0	0	1	48	28	29	9	3
4	5	0	0	19	214	10	18	0
5	24	0	16	6	6	708	16	58
6	7	0	16	11	19	38	95	1
7	5	4	20	3	0	103	3	323
	0	1	2	3	4	5	6	7

Predicted

Fig. 7: Confusion Matrix of detecting 8 shot types

IV. CONCLUSION

This study explores AI-driven techniques to analyze badminton match videos, aiming to automate rally segmentation, shot classification, and match event annotation. Throughout the development process, we employed various detection methods to extract initial data, yet inherent inaccuracies in the raw detections introduced biases in subsequent predictions. Although data correction techniques were applied to mitigate these errors, some deviations persisted, impacting final classification performance. Therefore, improving data extraction accuracy remains a key challenge in enhancing the overall precision of our system.

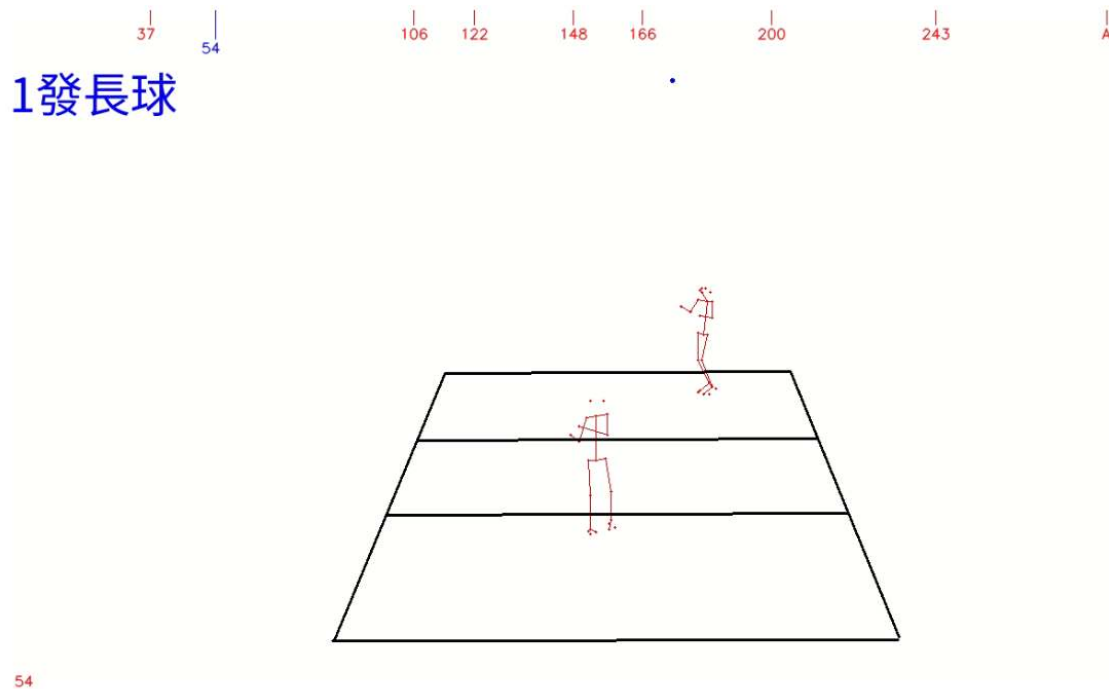
Despite these challenges, our final system successfully integrates detection and prediction results into annotated match videos, providing a structured and analytical way to view badminton gameplay. The outcomes demonstrate promising potential, yet there is room for further improvement. As a future direction, we aim to develop a web-based platform where users can upload their own match videos, have them automatically analyzed, and receive an annotated output for further study. This would enable players, coaches, and analysts to utilize AI-powered insights to review gameplay, study tactics, and refine training strategies. Ultimately, we hope our system will contribute to accessible, data-driven sports analysis, making advanced match evaluation tools available to a broader audience.

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(a) Match video segment displaying with court lines, body pose, shuttlecock, and hit detection along with detected shot type by our automated detection system.



(b) Displaying all the detected information solely without the video.

Fig. 8: Final result of integrating all detection based on the competition video. Here is an example of detecting the long service by player B. (b) provides a clearer way for users to see all the information detected from the given video. We also provide a demonstration system on our website for users to select which features they are interested in for displaying, with or without the original video.